

Thin-walled Pressure Vessels

Inner radius: C_i

Thickness: t , $C_i/t \geq 10$

Tube

$$\text{Hoop: } \sigma_h = + \frac{P C_i}{t}$$

$$\text{Axial: } \sigma_a = + \frac{P C_i}{2t}$$

Sphere

$$\sigma = \frac{P C_i}{2t} \text{ in All directions}$$

* 对算某一点的: $M, V, N, \dots$

仅 V_y / V_z 反, compared to 外力对该点的 Resultant Force / Moment

Combined Loading

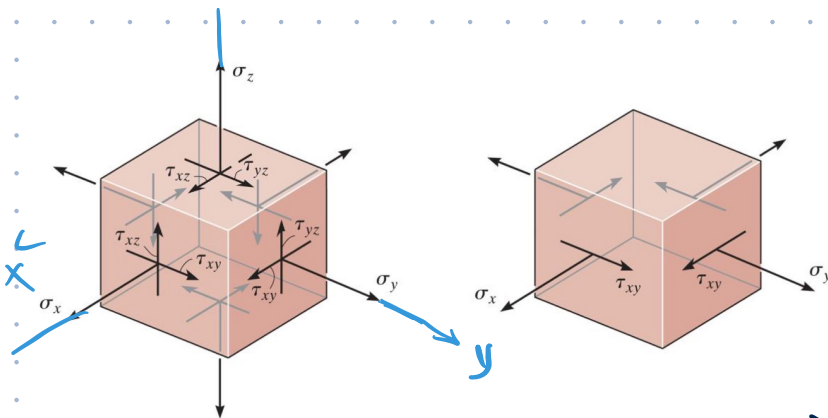
Normal Stress $\sigma_x, \sigma_y, \sigma_z$

Always Face Normal (向外)
⊕ Tensile ⊖ Compressive

Shear stress

τ_{xy} — Force Direction
Face Normal (向外)

⊕ x, y 同号
⊖ x, y 异号



Normal Force

$$\sigma = \pm \left| \frac{N}{A} \right|$$

⊕ Tensile
⊖ Compressive

Torsional Moment T_0

$$\tau = \pm \left| \frac{T_0 c}{J} \right|, \quad J = \begin{cases} \frac{\pi}{2} c^4 & \text{solid} \\ \frac{\pi}{2} (C_o^4 - C_i^4) & \text{Tube} \end{cases}$$

c : radius

Shear Force V

$$\tau = \pm \frac{VQ}{It}$$

$$I = \frac{1}{4} \pi c^4 \quad \text{Solid}$$

$$\frac{1}{4} \pi (c_o^4 - c_i^4) \quad \text{Tube}$$

$$Q = \sum \bar{c} A \quad c: \text{point} \rightarrow NA$$

Bending Moment

$$\sigma = \pm \frac{Mc}{I} \quad c: \text{point} \rightarrow NA$$

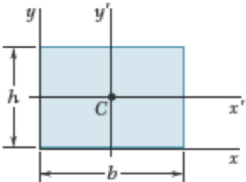
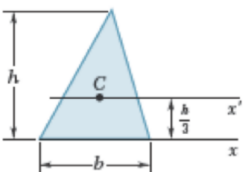
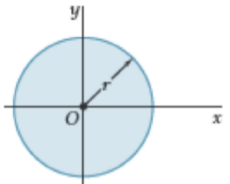
⊕ Tensile

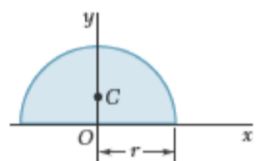
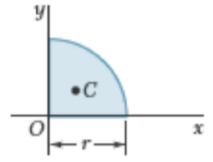
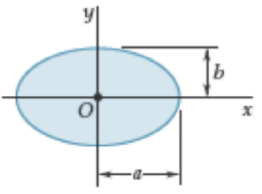
⊖ Compressive

$$\text{Hoop: } \sigma_h = + \frac{P c_i}{t}$$

Internal pressure

$$\text{Axial: } \sigma_a = + \frac{P c_i}{2t}$$

Rectangle		$\bar{I}_{x'} = \frac{1}{12} b h^3$ $\bar{I}_{y'} = \frac{1}{12} b^3 h$ $I_x = \frac{1}{3} b h^3$ $I_y = \frac{1}{3} b^3 h$ $J_C = \frac{1}{12} b h (b^2 + h^2)$
Triangle		$\bar{I}_{x'} = \frac{1}{36} b h^3$ $I_x = \frac{1}{12} b h^3$
Circle		$\bar{I}_x = \bar{I}_y = \frac{1}{4} \pi r^4$ $J_O = \frac{1}{2} \pi r^4$

Semicircle		$I_x = I_y = \frac{1}{8} \pi r^4$ $J_O = \frac{1}{4} \pi r^4$
Quarter circle		$I_x = I_y = \frac{1}{16} \pi r^4$ $J_O = \frac{1}{8} \pi r^4$
Ellipse		$\bar{I}_x = \frac{1}{4} \pi a b^3$ $\bar{I}_y = \frac{1}{4} \pi a^3 b$ $J_O = \frac{1}{4} \pi a b (a^2 + b^2)$

$$Q = \int y \, dA = \sum \bar{y} A$$

where $\bar{y}$: 小块块的 centroid to NA
 A : 小块块的面积

$$\Rightarrow \frac{\Delta H}{\Delta x} = \frac{M_D - M_C}{\Delta x I_z} Q \Rightarrow$$

$$\text{Shear Flow } q = \frac{VQ}{I_z} \text{ [N/m]}$$

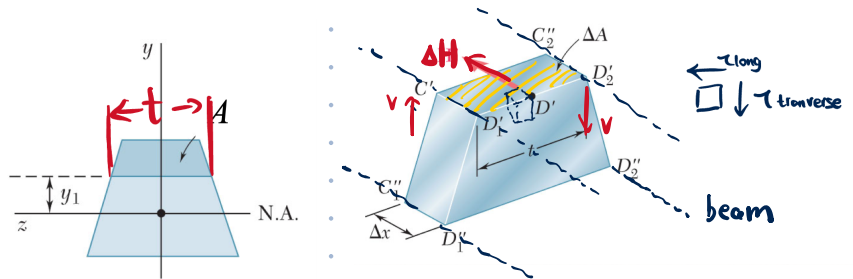
force per unit length (longitudinal)

$$\begin{cases} \frac{M_D - M_C}{\Delta x}, \frac{dM}{dx} = V \\ q = \frac{dH}{dx} \text{ shear flow} \end{cases}$$

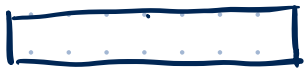
$$\text{Average Shear stress } \tau_{\text{long}} = \frac{q}{t} = \frac{VQ}{I_z t}$$

where t : 连接处长度 (cross-section)

τ_{long} along direction of t 连接处求均值



Built Up Beam With Fasteners



ΔH is Force longitudinal

$$dH = q \, dx$$

$$F_{\text{long}} = q \, \Delta x \rightarrow F_{\text{nail}}$$

$$F_{\text{nail}} = q \, \Delta S = \tau_{\text{nail}} A_{\text{nail}}$$

Where ΔS : Fastener 间距

τ_{nail} : 钉子的 shear stress (剪切)

A_{nail} : 钉子被截面积

* 注意 n 个钉子, $A_{\text{nail}} = \sum_n A$

此处求 q 技巧

① Q 要求被钉子分割的小块块